

Assignment_3

Student Name: 郭忠侑

1. Complete Exercise 9.4 in Hsieh' s book. Please visualize the data and/or results in some ways.

9.4

Analyse the dataset from the book website using principal component analysis (PCA). The dataset contains four time series x_1, x_2, x_3 and x_4 .

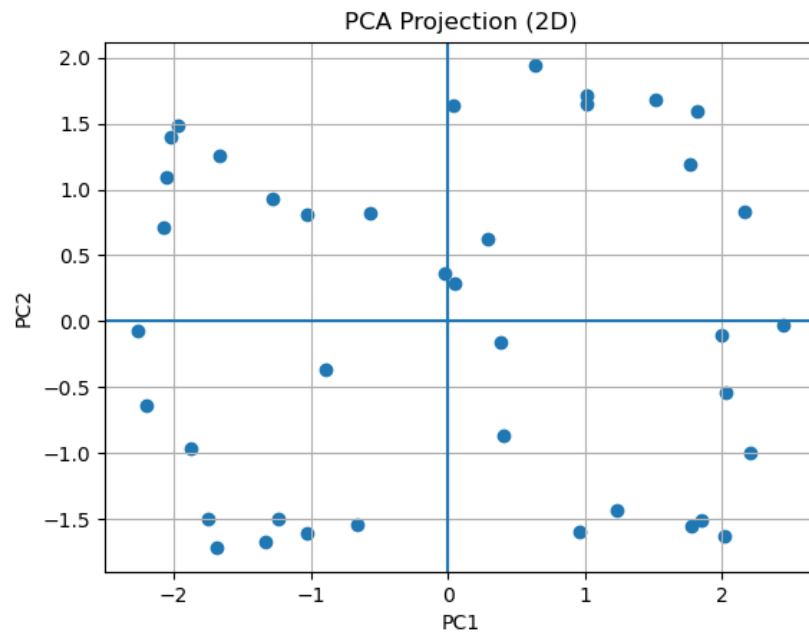
Figure 1: Problem1

I' ve used `sklearn` to do PCA in this assignment. The eigenvectors for PCA are

	PC1	PC2	PC3	PC4
x1	0.513685	-0.469876	0.682832	0.221550
x2	0.405938	0.627278	-0.087502	0.658847
x3	-0.553820	-0.409653	-0.098287	0.718198
x4	-0.514417	0.466823	0.718626	-0.032063

The explained variances by each PCs are: [0.59785561 0.37827087 0.01810953 0.005764].

It' s evident that we can reduce the dimension of data down to 2, and then plot the scatter plot as below:



[Problem1 Code](#)

2. Complete Exercise 9.5 in Hsieh' s book. Please visualize the data and/or results in some ways.

9.5

From the given dataset containing observations of chlorophyll concentration (chl), dissolved oxygen (dox), suspended particulate matter (spm), salinity (sal) and water temperature (temp) in San Francisco Bay, standardize the five variables and perform (a) principal component analysis (PCA). Compare the PCA eigenvectors with those from (b) E-frame varimax rotated PCA (RPCA) and (c) A-frame varimax RPCA for the first k modes, using four different k values ($k = 2, 3, 4$ and 5). Also compare the percentage of variance explained by the leading k modes in (a), (b) and (c).

Figure 2: Problem2

Algorithms for solving A-frame/E-frame varimax

For the varimax(take E-frame for example), the textbook only gave criteria but without explicit algorithm:

$$R_{\text{varimax}} = \arg \max_R L(R; A) = \arg \max_R \sum_{j=1}^k \left\{ \sum_{i=1}^n (\tilde{A}_{ij})^4 - \frac{1}{n} \left(\sum_{i=1}^n (\tilde{A}_{ij})^2 \right)^2 \right\}$$

where $\tilde{A}_{ij}$ is the element of the matrix $\tilde{A} = AR$, and k is the number of dimension we choose, in which I have always follow the notation of the textbook as much as possible.

By asking ChatGPT, it gave me an algorithm for varimax as follows:

1. Initialize $R_0 = I_k$
2. compute $\nabla_R L(R; A) = P(R; A)$, where $P(R; A)$ can be derived as matrix representation:

$$P(R; A) = A^T \left[\tilde{A}^{3(\text{ele})} - \frac{1}{\# \text{ rows of } A} \tilde{A}(\text{diag}(\tilde{A}^T \tilde{A})) \right]$$

$\tilde{A}^{3(\text{ele})}$ means element wise cubed

$\text{diag}(M)$ means the diagonal terms of M and keep the dimension of M , for example

$$\text{diag} \left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

3. do svd of $P = U\Sigma V^T$
4. update R by $R = UV^T$
5. repeat until $\text{trace}(\Sigma)$ makes little progress

In fact, this algorithm is also used by [sklearn_ortho_rotation function](#).

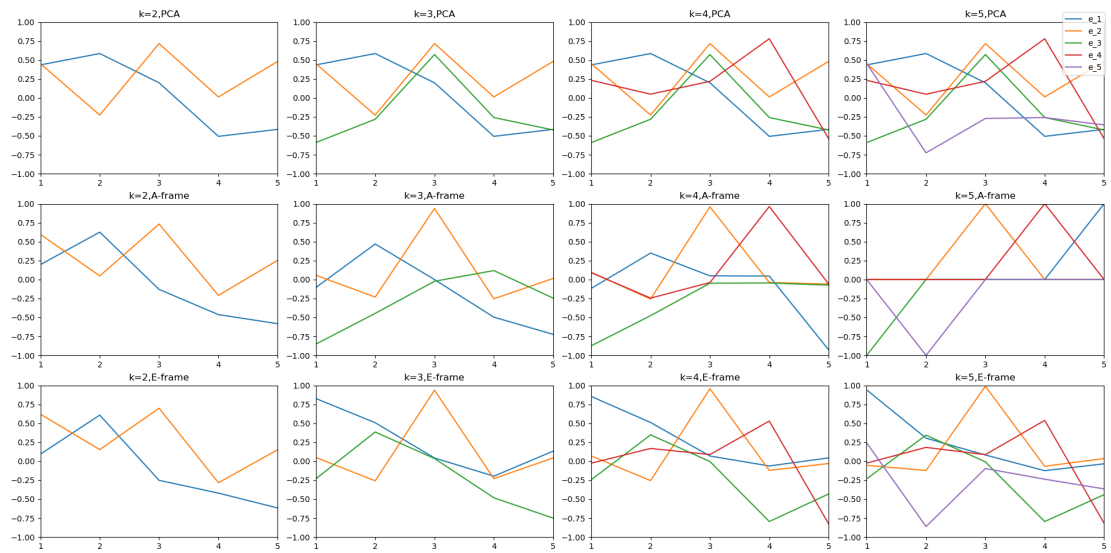
I will include some more detailed derivation on this in other notes.

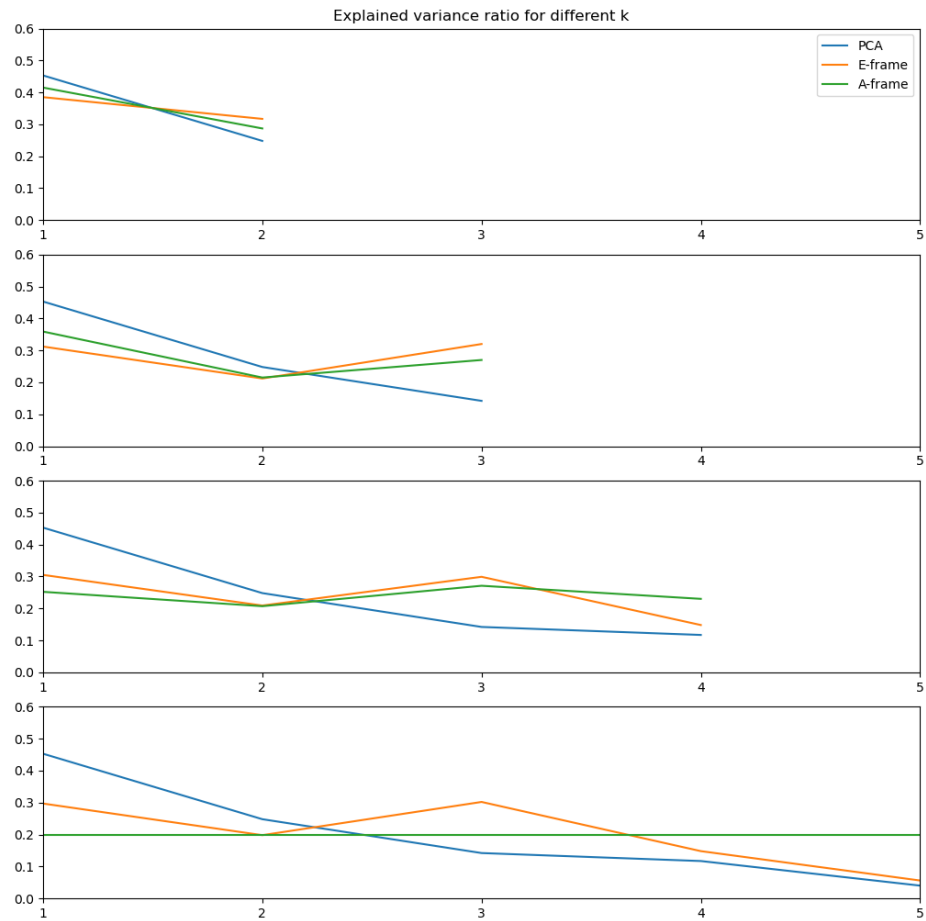
Explained variance ratio for different k

Once we've obtained the varimax rotation matrix R , compute $\tilde{A} = AR$ and then $\tilde{S} = (\tilde{A}^T A)/(n - 1)$, the explained variances along the new axis are the diagonal terms in $\tilde{S}$, namely $\text{diag}(\tilde{S})$, and here's the results:

```
k=2
PCA_variance=[0.453 0.248]
E-frame_variance=[0.385 0.317]
A-frame_variance=[0.415 0.287]
k=3
PCA_variance=[0.453 0.248 0.142]
E-frame_variance=[0.312 0.212 0.32 ]
A-frame_variance=[0.359 0.215 0.27 ]
k=4
PCA_variance=[0.453 0.248 0.142 0.117]
E-frame_variance=[0.305 0.209 0.299 0.148]
A-frame_variance=[0.252 0.207 0.271 0.23 ]
k=5
PCA_variance=[0.453 0.248 0.142 0.117 0.04 ]
E-frame_variance=[0.297 0.198 0.302 0.148 0.056]
A-frame_variance=[0.2 0.2 0.2 0.2 0.2]
```

Some plots





Some discussion

- The ordering relation for explained variances is not always preserved. Take $k=2$ for example, $\text{var1}(\text{pca}) > \text{var1}(\text{e-frame})$, but $\text{var2}(\text{pca}) < \text{var2}(\text{e-frame})$.
- For A-frame rotation, as its objective function behaves, it makes some components of column vectors in $\tilde{E}$ to be zero, as can be clearly seen from the case of $k=5$ (A-frame) diagram.
- The components plot varies with k for the rotated PCA, while it does not for the ordinary PCA, since for different k , it will introduce different variance effect into the objective function of the varimax.

[Problem2 Code](#)